# TECH OFFER FORM

## General Guidelines

Kindly take note of the following details before filling up your Tech Offer:

- The information provided should not contain the following information: owner's identity, trademarks, brand names, etc., and be written in the third-person point of view.
- Images provided should be representative of the technology and should not contain information that could potentially lead to the identification of the technology owner.
- The technology will be kept agnostic for IPI to assist in filtering unrelated queries.
- Sections indicated with * are compulsory fields

## Information for dissemination

**Title (capitalise each new word) ***

Climate Optimization for Indoor Farming

**Technology Overview (50 - 250 words) ***

Indoor farmers intuitively know that the environment can affect the growth of the crop. The indoor climate can either contribute to the yields or, in unfortunate circumstances, lead to loss of the crop.

However, it is not always easy to create an ideal environment for the crop. At the moment, most building management systems (BMS) are closed off and don't talk with operational systems, such as the ones used for indoor farming. Moreover, existing smart building solutions usually focus on energy efficiency alone and overlook the need to balance energy savings with crop growth, operational efficiency, low emissions, and other factors.

Our technology overcomes this challenge by allowing the indoor farmer to input the optimal conditions for the crop and letting the system optimize the rest. The user only has to specify which crop(s) they are growing or, if they prefer, what 'recipe' the system should follow. Accuracy can be improved by adding environmental sensors.

We are looking for new or established indoor farms interested in piloting this solution in Singapore or APAC.

*A brief summary or description of the invention - What is the problem solved by this technology? Who is going to buy this technology? How is this IP addressing a need/problem in the marketplace? Note: Please do not mention your company or brand in any of the descriptions.*

**Keywords**

**Technology Readiness Level (TRL) ***

6

*Refer to* ***- Technology Readiness Level Reference Table*** *for details.*

**Technical Features & Specifications (50 - 250 words) ***

The technology offered is a that optimizes building performance in where plants are grown (i.e., indoor farms). What makes it unique is the machine learning techniques it uses to find the optimal environment. Customers in the indoor farming market can expect energy savings of 25% to 35%, improved crop yields by 25% to 75%, and 25% reduction in man-hours.

*What does the technology consist of? E.g. Video camera, software algorithm*
*Who is/are the ideal collaboration partner(s) in the value chain? E.g. medical institutions, device manufacturers*

**Potential Applications (20 - 250 words) ***

This technology is directly applicable to the indoor farming market in APAC and globally. The market size in APAC alone is USD 1.2B.

*In which industry can this technology be deployed?*
*What are the products that can be marketed based on this technology?*

**Market Trends & Opportunities (optional)**

Opportunity to extend the functionality to other aspects of vertical farming in addition to climate management through BMS. We can look into soil conditions as well as plant video analytics.

*What is the approximate market size for this technology?*
*What makes this IP attractive to the market and what are its unique selling points against similar technologies in the market.*

**Unique Value Proposition (20 - 250 words) ***

BMS provider to advise them on how to better manage their facility for [illegible] er, farmers need more than energy conservation. They need actionable insights on whether expending more energy might actually benefit their crops and their business. This is more than an operational challenge as it requires looking at data across crop, technology, and enterprise (business performance). Farmers may want to do it themselves, but they'll quickly find that the data analysis expense can cost them hundreds of thousands of dollars and take many months to complete. Ambient gives them the information they need in three simple clicks using data they already likely have.

*How is this technology an improvement over the current "State-of-the-Art"?*
*What is the Unique Value Proposition (UVP) in comparison to the current "State-of-the-Art"?*

**Competitors** *

Our main competitors are the AI platform providers who can customize their solution to the customer's needs. However, this is very expensive and prohibitive to most vertical farms that are not yet profitable. Doing it yourself is also an alternative, but the farmers may end up investing in capabilities that they can't justify financially. For example, to deploy a single AI application end-to-end may require three highly skilled teams across software development, data engineering, and data science (AI and ML modelling). This is not only hard to assemble, it is also very expensive, and takes a long time. Finally, another competitor could be the hardware supplier for vertical farms. These are the companies that manufacture or procure various components such as lights, racks, and even the building management system (BMS). They are good in hardware, but often rely on outsourcing their software development to 3rd party vendors, leading to quality, security, and cost overrun issues that the end-user (the farmer) is not aware of until the system is 'handed-over'. Then they may realize they need to pay more than expected to maintain it, which may not provide the ROI needed for justifiable technology investments.

*Indicate the top 3 competitors of your technology and state how are they different.*

**Technology Categories** *

Software (SaaS), Data Analytics, Decarbonization, AgriCulture, Vertical Farming

*E.g. Chemicals; Electronics; Energy; Environment; Clean Air & Water; Foods; Green Building; Healthcare; Infocomm; Life Sciences; Logistics; Manufacturing; Materials; Personal Care; Waste Management & Recycling*

**Ideal Collaboration Partners** *

We'd love to collaborate with a notable vertical farm operator to test bed the solution, deploy it across their facility, and co-develop further applications (in addition to climate optimization). We can provide them with all the software capabilities needed at the fraction of the cost than anyone else, while providing world-class quality and service. We're also open to licensing or jointly developing further additions to the software through an industry coalition of partners. This way the industry saves money while getting the digital capabilities they need and delivering even bigger quantities of high-quality produce that is need by Singapore and beyond.

*Provide information about the type of collaboration partners you'd like to work with e.g. Institute of Higher Learning (IHLs), food testing laboratories, wafer fabrication partners, deep-tech companies with data analytics capabilities*

**Collaboration Mode** *

| | | |
|---|---|---|
| X R&D Collaboration | X Licensing | ☐ IP Acquisition |
| X Test-Bedding | | |

## Others

Please also provide the following materials for your Tech Offer upon submission of this form:

i) **Representative Image(s)**

*Please provide at least two representative images (landscape: 1000 x 540 pixels, square: 400 x 400 pixels; or higher) that best describes the technology.*

ii) **Attachments* (if any)**

*Any additional non-confidential information or attachments that will assist in understanding the technology in greater detail, e.g. technology brochure, related test reports, patent document, journal articles, etc.*

** Note: These will be used solely for the purpose of understanding the technology and developing an accurate write-up on this technology.*

## For internal use and reference only (Not published):

**Organisation Information**

| | | | |
|---|---|---|---|
| Organisation Name: | Ambient Systems Pte. Ltd. | | |
| Contact Person: | Ivan Damnjanovic | | |
| Email: | ivan@ambientsystems.ai | Phone: | [redacted] |
| **IP Protection**: | Singapore patent-pending (10202105064T) | | |

*E.g. Patented, Patent Application in Progress, Trade Secret, etc. Please list registered patent application and granted patent numbers if applicable.*

**Declaration**

To facilitate the conduct of the Services by IPI, the Company agrees that IPI may disclose the Company's Tech Offer to:

☐ IPI's technology partners to enable IPI to perform the Services, and where necessary, the IPI shall notify such technology partners of the need to execute a separate confidentiality agreement with the Company.

☐ Publish the Tech Offer on IPI's web portal, newsletter and editorials so that IPI can leverage its network to help find suitable partners for the Company; however such publication shall first be anonymised such that the Company's name is not disclosed in connection with the Tech Offer.

*Please tick the option(s) that applies.*

## : Technology Readiness Level (TRL) Reference Table

| TRL | Physical Sciences & Engr | Healthcare (Pharmaceutical) | Healthcare (Medtech) | Healthcare (Diagnostics) | Simplified |
|---|---|---|---|---|---|
| 1 | Basic principles observed | Basic principles observed | Basic principles observed | Basic principles observed | Proof-of-Concept |
| 2 | Technology concept formulated | Technology concept formulated | Technology concept formulated | Technology concept formulated | Proof-of-Concept |
| 3 | Experimental proof of concept | Experimental proof of concept in vitro and in vivo research models | Experimental proof of concept in vitro and in vivo research models | Experimental proof of concept in vitro | Proof-of-Concept |
| 4 | Technology validated in lab | Proof of concept demonstrated in defined laboratory/animal models | Proof of concept demonstrated in defined laboratory/animal models | Analytical validation | Prototype in Lab |
| 5 | Technology validated in relevant environment | Non-clinical and pre-clinical research studies, & initial demonstration of feasibility and efficacy | Product Development Plan | | |
| 6 | Technology demonstrated in relevant environment | Phase 1 clinical trials | Phase 1 clinical trials | | |
| 7 | System prototype demonstration in operational environment | Phase 2 clinical trials | Clinical safety and effectiveness trials in operational environment | Clinical validation in 1 site | Prototype in Live Environment |
| 8 | System complete and qualified | Phase 3 clinical trials | Overall risk-benefit Trials | | |
| 9 | Actual system proven in operational environment | Pharmaceutical can be distributed or marketed | Medical device can be distributed or marketed | Clinical validation in multi-site | Ready-to-Market |